Software Test Plan (STP)

Traffic Management System (Simulation)

Field	Value
Project	Traffic Management System (Simulation)
Course	Software Engineering UE24CS341A
Institution	PESU
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Authors / Team

#	Name	SRN	Primary responsibility
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Revision history

Version	Date	Author	Change summary	Approval
0.2	2026-09-07	Venkata Srivaibhav Nukaraju; V Vyas; Vikas N Naik; Vikhyath Bharadwaj K S	Finalized simplified SRS for implementation and review.	Pending
0.3	2026-09-14	Venkata Srivaibhav Nukaraju; V Vyas; Vikas N Naik; Vikhyath Bharadwaj K S	Created Software Architecture and Design Specification (SAD) for Traffic Management System (Simulation).	Pending
1.0	2026-09-15	Venkata Srivaibhav Nukaraju; V Vyas; Vikas N Naik; Vikhyath Bharadwaj K S	Created Software Test Plan (STP) for Traffic Management System (Simulation) based on the finalized SRS and SAD.	Pending

1. Introduction

Purpose: This document defines a simple test plan for Traffic Management System (Simulation) (TMS-Sim). It identifies what will be tested, the basic test strategy, environment, schedule, deliverables, responsibilities, risks, traceability, and test metrics.

Scope: Testing covers the software-only desktop application described in the SRS: road-network modelling, fixed-time traffic-signal configuration, traffic-demand generation, time-stepped microscopic simulation, simulation controls, 2D visualization, metrics and reporting, comparison/export of completed runs, and the local scenario library.

Out of scope: Real-world traffic signals, sensors, field hardware, 3D visualization, distributed/multi-machine simulation, advanced lane changing, calibration against field data, public transport scheduling, and pedestrian/cyclist micro-simulation.

References: Traffic Management System (Simulation) SRS v0.2, Software Architecture and Design Specification, and UML system models/diagrams.

2. Test Items

- Road Network Modelling
- Traffic-Signal Configuration
- Traffic Demand Generation
- Simulation Execution & Control
- 2D Visualization
- Metrics, Results & Reporting
- Scenario Library

3. Features to be Tested

The following SRS requirements and acceptance-test references are the main test items:

- TMS-F-010 / TC-NET-01 — Create/edit a directed road network and verify saved network attributes reload correctly.
- TMS-F-011 / TC-NET-02 — Define intersection control type and permitted turning movements.
- TMS-F-012 / TC-NET-03 — Validate the network and block a run when seeded network defects exist.
- TMS-F-013 / TC-NET-04 — Import/export a complete version-tagged JSON scenario.
- TMS-F-020 / TC-SIG-01 — Define fixed-time signal phases and timing values.
- TMS-F-022 / TC-SIG-02 — Reject invalid or conflicting signal plans.
- TMS-F-030 / TC-DEM-01 — Configure entry-point arrival rates and turning ratios.
- TMS-F-032 / TC-DEM-02 — Verify same scenario + seed produces identical vehicle arrivals and metrics.
- TMS-F-040 to TMS-F-043 / TC-SIM-01 to TC-SIM-04 — Run the time-stepped simulation, use controls, stop at configured duration, and verify the 2D view.
- TMS-F-050 to TMS-F-054 / TC-MET-01 to TC-MET-05 — Compute/display metrics, persist completed runs, compare runs, and export CSV results.
- TMS-F-060 / TC-LIB-01 — Save, list, load, rename, and delete scenarios.
- TMS-NF-001 / TC-PERF-01 — Run the reference scenario with up to 10 intersections and 500 vehicles while maintaining at least 15 FPS during normal visualization.
- TMS-NF-002 / TC-REL-01 — Reliability validation.
- TMS-NF-003 / TC-UX-01 — Usability validation.
- TMS-NF-004 / TC-PORT-01 — Windows 10/11 portability validation.

4. Features Not to be Tested

- Real-world traffic signals, sensors, controllers, and field hardware.
- 3D visualization and distributed/multi-machine simulation.
- Advanced lane-changing behaviour.
- Calibration against real field-measured traffic data.
- Public transport scheduling and pedestrian/cyclist micro-simulation.

5. Test Approach / Strategy

Levels:

- Unit testing — individual simulation, network, demand, metrics, storage, and UI modules.
- Integration testing — interactions among the major software components and scenario storage.
- System testing — complete analyst workflows from configuration through simulation and results.
- Acceptance testing — verify the SRS acceptance criteria for retained requirements.

Types:

- Functional testing.
- Regression testing.
- Performance testing.
- Usability testing.
- Reproducibility testing using the same scenario and PRNG seed.

Entry Criteria: A stable implementation build is available, required test scenarios/data are prepared, and the Windows test environment is ready.

Exit Criteria: Planned test cases are executed; High-priority functional requirements pass their acceptance tests; all four NFRs are checked; TMS-F-032 reproducibility passes; the RTM has test links; and no Critical or High defect remains open.

5.1 Security Validation

The SRS is a local software-only application and does not require network communication. Security validation will therefore focus on safe local handling of scenarios and input validation:

- Verify invalid network, signal, demand, and scenario inputs are rejected safely.
- Verify imported scenario files are validated before use.
- Verify scenario/run files are accessed through the application's supported storage operations.
- Verify no unintended network communication is required for normal operation.

6. Test Environment

Hardware: Reference hardware from the SRS — 4-core Intel i5-11400H @ 2.70 GHz, 16 GB RAM; standard display, keyboard, and pointing device.

Software: Windows 10/11; Python 3.11 + PySide6 implementation environment; SQLite database or scenario folder on local disk, as applicable to the implementation.

Tools: Unit/integration test framework used by the development team, Jira for issue tracking, and CSV/JSON files for scenario and result validation.

Test Data: Sample road networks, signal plans, arrival rates, turning ratios, valid/invalid configurations, seeded scenarios, and completed simulation runs.

7. Test Schedule

Milestone	Date
Test case preparation	16-Sep-2026
Test environment setup	17-Sep-2026
Test execution start	18-Sep-2026
Test execution end	24-Sep-2026
Acceptance testing	25-Sep-2026 to 26-Sep-2026

8. Test Deliverables

- Software Test Plan
- Test Cases
- Test Data / Sample Scenarios
- Test Execution Results
- Defect Reports
- Requirements Traceability Matrix updates
- Test Summary Report

9. Roles and Responsibilities

Role / Member	Responsibility
Venkata Srivaibhav Nukaraju	Simulation Engine & core model; support simulation test execution.
V Vyas	Network & Signal-plan modelling; support network and signal tests.
Vikas N Naik	UI, 2D Visualization & UX; support visualization/usability tests.
Vikhyath Bharadwaj K S	Metrics, Reporting, QA & Test; coordinate test cases, evidence, defect logging, and reporting.

10. Risks and Mitigation

Risk	Mitigation
Simulation defects affect system testing	Run unit/integration tests first and use small sample scenarios for debugging.
Large scenarios reduce visualization performance	Use the SRS reference scenario and measure FPS against the 15 FPS target.
Invalid network or signal configuration	Maintain positive and negative test cases based directly on SRS validation rules.
Scenario data is lost or corrupted	Keep controlled test copies and verify save/load/import/export operations.

11. Assumptions & Dependencies

- The implementation follows the simplified SRS v0.2.
- Users can supply their own network data or use bundled sample scenarios.
- Required Python libraries are available under permissive open-source licenses.
- Testing is performed on Windows 10/11.
- Correctness is evaluated against the documented model equations, not field measurements.

12. Suspension & Resumption Criteria

Suspend testing if the application cannot start, the environment is unavailable, or a blocking defect prevents execution of a significant portion of planned tests.

Resume testing after the blocking defect is fixed, a stable build is available, and the test environment/data are restored.

13. Test Case Management & Traceability

The SRS already provides a test-case reference for each retained functional and non-functional requirement. The test team will maintain the RTM by recording execution status and evidence.

SRS Req ID	Test Case	Area
TMS-F-010	TC-NET-01	Network Modelling
TMS-F-012	TC-NET-03	Network Validation
TMS-F-020	TC-SIG-01	Signal Configuration
TMS-F-022	TC-SIG-02	Signal Validation
TMS-F-030	TC-DEM-01	Demand Generation
TMS-F-032	TC-DEM-02	Reproducibility
TMS-F-040	TC-SIM-01	Simulation Engine
TMS-F-041	TC-SIM-02	Simulation Controls
TMS-F-043	TC-SIM-04	2D Visualization
TMS-F-050	TC-MET-01	Metrics
TMS-F-053	TC-MET-04	Run Comparison
TMS-F-054	TC-MET-05	CSV Export
TMS-F-060	TC-LIB-01	Scenario Library
TMS-NF-001	TC-PERF-01	Performance
TMS-NF-002	TC-REL-01	Reliability
TMS-NF-003	TC-UX-01	Usability
TMS-NF-004	TC-PORT-01	Portability

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed / blocked
- Requirement coverage
- Defect count by severity
- Defect aging
- Performance result in FPS for the reference scenario
- Reproducibility pass/fail result

Reports: Test execution status, defect status, RTM status, and final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
Course Coordinator		